

Generation Interconnection System Impact Study Report

For Queue Project Q22-2222

Kettner 100 kV
February 26, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2222. The Interconnection Customer proposes a 141.7 MW solar photovoltaic facility interconnecting at the Kettner 100 kV substation in the Indiana area. The request was received on November 07, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 637 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,055,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Gates – Sneath-Pacifica Tap 100 kV Line
Contingency: Gates – Sneath-Pacifica Tap 100 kV #2
Loading: 159.6% (Rating: 147 MVA)
DFAX: 76.6%
Pre-project loading: 85.7%
2. Facility: Gates – Sneath-Pacifica Tap 100 kV Circuit 2
Contingency: Gates – Sneath-Pacifica Tap 100 kV Line 1
Loading: 159.6% (Rating: 147 MVA)
DFAX: 76.6%
Pre-project loading: 85.7%
3. Facility: Dunlap – Sneath-Pacifica Tap 100 kV
Contingency: Stubey Road – Idle Hour 100 kV Line
Loading: 131.5% (Rating: 200 MVA)
DFAX: 62.7%
Pre-project loading: 87.0%
4. Facility: Lusher – Sneath-Pacifica Tap 100 kV
Contingency: Sneath-Pacifica Tap – Dunlap 100 kV Line
Loading: 101.9% (Rating: 194 MVA)
DFAX: 74.2%
Pre-project loading: 47.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gates – Sneath-Pacific Tap 100 kV #1	Rebuild line with higher-capacity conductor	\$1,928,000	31% (\$598,000)
Gates – Sneath-Pacific Tap 100 kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,758,000	31% (\$545,000)
Dunlap – Sneath-Pacific Tap 100 kV	Rebuild line with higher-capacity conductor	\$1,734,000	26% (\$444,000)
Lsher – Sneath-Pacific Tap 100 kV Line	Reconductor line	\$1,510,000	31% (\$468,000)
	Total allocated to project		\$2,055,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.